

Code-Layout, Readability and Reusability

Date	2 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction Using Machine Learning
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the SmartLender project's code quality in terms of layout, readability, maintainability, and reusability. The project follows a modular structure with separate folders for templates, static files, machine learning models, and Flask backend code. Proper naming conventions, comments, and reusable functions make the application easy to understand, maintain, and extend.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Used 4-space indentation throughout the project.
2	Proper File Structure	Files and folders are logically organized	Yes	Project organized into data, n
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables like credit_df, application_df, model, prediction.
4	Function / Method Names	Functions are descriptively named	Yes	Functions have descriptive names such as preprocess_data() and predict_approval()
5	Code Comments	Inline and block comments explain logic	Yes	Important sections are documented with
6	Modular Design	Code is split into reusable functions/modules	Yes	Separate modules for preprocessing, model training, and prediction.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate code removed using reusable functions
8		Error Handling		

	Error Handling	Exceptions and errors are handled gracefully	Yes	Basic validation and exception handling implemented.
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Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing Module	Python, pandas	Training & Prediction	High
2	Label Encoding Module	Scikit-learn	Data Cleaning	High
3	Model Prediction Module	Scikit-learn	Flask Application	High
4	Flask Routes	Flask	Web Application	Medium
5	Utility Functions	Python	Entire Project	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized folder structure with separate modules.
Readability	5	Clear variable names and consistent coding style.
Reusability	5	Reusable preprocessing and prediction modules.
Documentation / Comments	4	Key sections documented with comments.
Overall Score	4.8 / 5	Clean, modular, and maintainable code suitable for academic projects.